

TECHNOLOGY EXPLORER SERIES

BOOK 2

Inside a Smartphone

The Tiny World in Your Hand



Fansci Solutions

Inside a Smartphone: The Tiny World in Your Hand

The Secret Life of Technology, Book 2

© 2026 Fansci Solutions

All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior written permission of the publisher.

Published by Fansci Solutions

Series: The Secret Life of Technology

Book 1: The Secret Life of the Internet

Book 2: Inside a Smartphone: The Tiny World in Your Hand

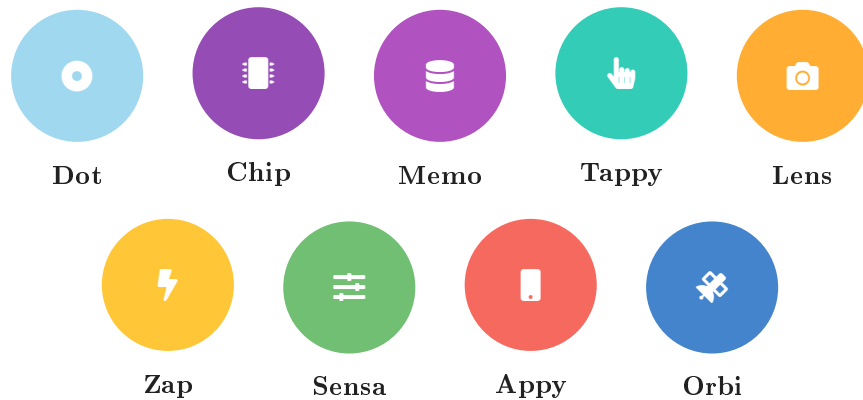
Designed for readers ages 8–12.

Illustrations and diagrams produced with TikZ.

Typeset in Source Sans Pro using L^AT_EX.

Meet Your Guides

Welcome back, explorer! Some old friends are here, and there are exciting new ones to meet. They will guide you through every corner of the tiny computer in your pocket.



Dot	Returning from Book 1 — a glowing packet of information, your main guide.
Chip	A tiny silicon processor — the brain of the phone, always thinking.
Memo	A cheerful memory card who never forgets a thing.
Tappy	A lively fingertip who brings the touchscreen to life.
Lens	A clever little camera lens who loves capturing moments.
Zap	A buzzing battery cell, full of energy and ready to power up.
Sensa	A curious sensor cluster who feels every tilt and twist.
Appy	A busy worker bee who runs all your favourite apps.
Orbi	A satellite orbiting high above Earth, helping phones find their way.

Welcome

Last time we explored the Internet — the giant network that connects the world. Now it is time to open up the tiny computer you carry in your pocket! We will shrink down and travel inside a smartphone, meeting the amazing parts that work together to take photos, play music, find your location, and keep you connected. Are you ready? Follow Dot inside!

Contents

Meet Your Guides	iii
Welcome	iii
1 What Is a Smartphone?	1
2 Meet the Brain: The Processor	5
3 Memory and Storage	9
4 The Amazing Touchscreen	12
5 Tiny Cameras, Big Pictures	16
6 The Battery Power Station	20
7 The Hidden Sensors	24
8 Apps: The Workers Inside Your Phone	28
9 GPS: Finding Your Way	32
10 Stay Safe and Smart	36
Big Ideas to Remember	40
Glossary	41
Quiz Time!	44
Parent & Teacher Guide	48

Chapter 1

What Is a Smartphone?

You hold it in your hand every day. But have you ever wondered what is going on inside?

Mission Start: Shrink Down and Step Inside!

Dot Returns

Sam was sitting on the steps outside, holding a smartphone.

The screen glowed. Music was playing. A photo notification popped up. Then a map. Then a game.

Sam frowned and said:

“This little thing does so much. How does it all fit inside?”

A tiny glowing dot zoomed out from the corner of the screen with a familiar grin.

Dot

“Remember me? Last time we explored the Internet together. Now there is a whole new adventure waiting — inside this very phone. Ready? Follow me in!”

More Than Just a Phone

The word “smartphone” has the word *phone* in it — but a smartphone is so much more.



Imagine This

Imagine a Swiss Army knife. It looks like a simple knife, but unfold it and you find scissors, a screwdriver, a saw, a magnifying glass, and more. A smartphone is like that — it folds dozens of devices into one thin rectangle.

Inside one smartphone you will find:

- A **telephone** (calling and texting)
- A **camera** (photos and videos)
- A **music player**
- A **navigation device** (maps and GPS)
- A **web browser** (for exploring websites on the internet)
- A **calculator**, **torch**, **calendar**, and much more



Did You Know?

The first iPhone was released in 2007 — about 20 years ago. Before that, phones could only call and send text messages.

A Computer That Fits in Your Pocket

Here is the most surprising fact: a smartphone is a **computer**. Not just similar to a computer — it *is* one. Any computer does three big jobs: it takes in information, works on it, and gives you a result. Your smartphone does all three, all day long.

Like a laptop or desktop, a smartphone has:

- A **processor** (the brain that does all the thinking)
- **Memory** (for working on things right now)
- **Storage** (for keeping things long term)
- **Input** (touchscreen, microphone, camera)
- **Output** (screen, speaker, vibration)

Quick Map

How powerful is a smartphone?

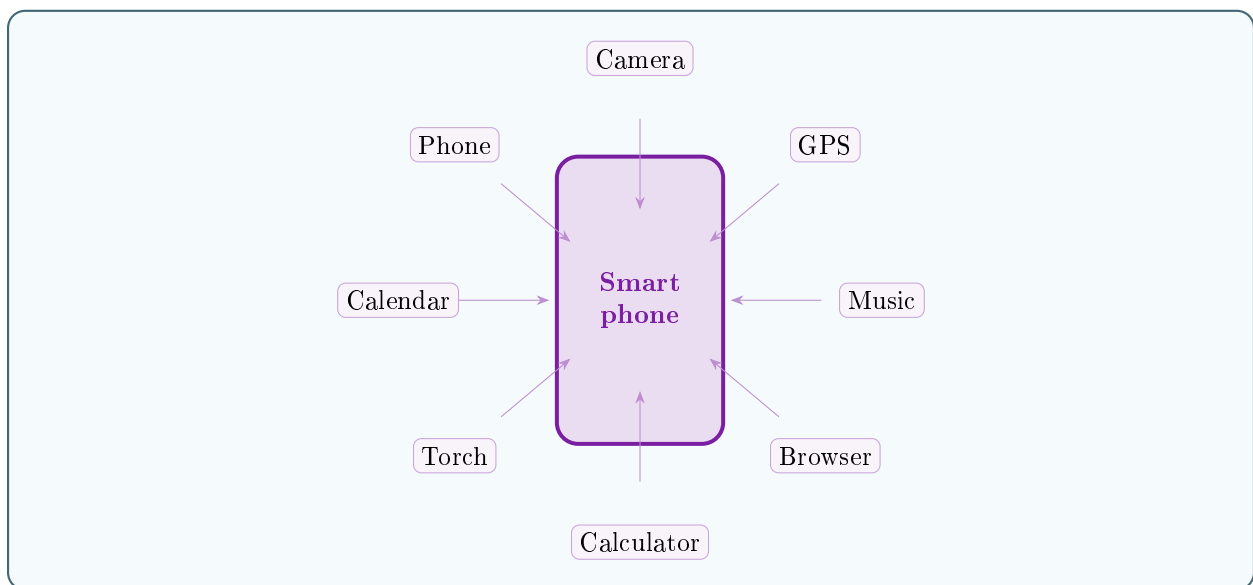
1. A 1990s desktop computer could do about 100 million calculations per second.
2. A modern smartphone processor can do over 15 *billion* steps per second.
3. That is more than 150 times faster — and it fits in your pocket.

How Smartphones Changed Everyday Life

Before smartphones, if you wanted to:

- Check the weather — you watched TV or bought a newspaper.
- Take a photo — you used a separate camera with film.
- Find directions — you read a paper map.
- Listen to music — you carried a separate music player.

Today, one device in your pocket does all of these things instantly.



The many tools packed into one smartphone.



New Words

Smartphone Processor

A mobile phone that is also a computer, camera, map, and much more.
The brain of a device — carries out instructions. (We meet it in Chapter 2.)

Memory Storage

Holds information the device is working with right now.
Keeps photos, apps, and files even when the phone is off.

Input

The ways information goes *into* a phone — touch, voice, camera.

Output

The ways information comes *out* — screen, sound, vibration.



Try It

Count the devices in your phone!

Look at the home screen of a smartphone (with a grown-up nearby). Find an app for each of these:

- A clock or alarm
- A camera
- A weather app
- A music or video player
- A map

How many did you find? Could you find even more?



Big Idea

A smartphone is a **tiny computer** that combines dozens of devices into one. Inside, it has a brain, memory, storage, cameras, and sensors all working together.

Dot

“Now that you know what a smartphone is, let’s meet the most important part inside it — the brain. Follow me to Chapter 2!”

Chapter 2

Meet the Brain: The Processor

Everything your phone does — every tap, swipe, photo, and song — starts with one tiny chip doing billions of calculations every second.

Mission: Follow an Instruction Through the Brain!

Meet Chip

Dot led Sam deeper into the phone, past glowing wires and humming circuits, until they reached the heart of the device — a flat, shiny square no bigger than a fingernail.

A tiny character with sparkling eyes and a crisp silver jacket waved at them.

Chip

“Welcome! I am Chip — the processor. Everything you do on this phone passes through me first. When you tap the screen, I get the message. When you take a photo, I handle it. Even reading this book on a screen would go through me.”

What Does a Processor Do?

A **processor** (also called a CPU — Central Processing Unit) is the brain of the smartphone. Its job is to follow **instructions**.

Every action you take on a phone is turned into a long list of instructions. The processor reads each instruction and carries it out, one after another, billions of times every second.



Imagine This

The processor is like a chef in a very busy kitchen. Orders keep arriving: chop this, heat that, plate this next. The chef follows each instruction in order, very fast. The processor does the same thing — except it handles billions of instructions in the time it takes you to blink.

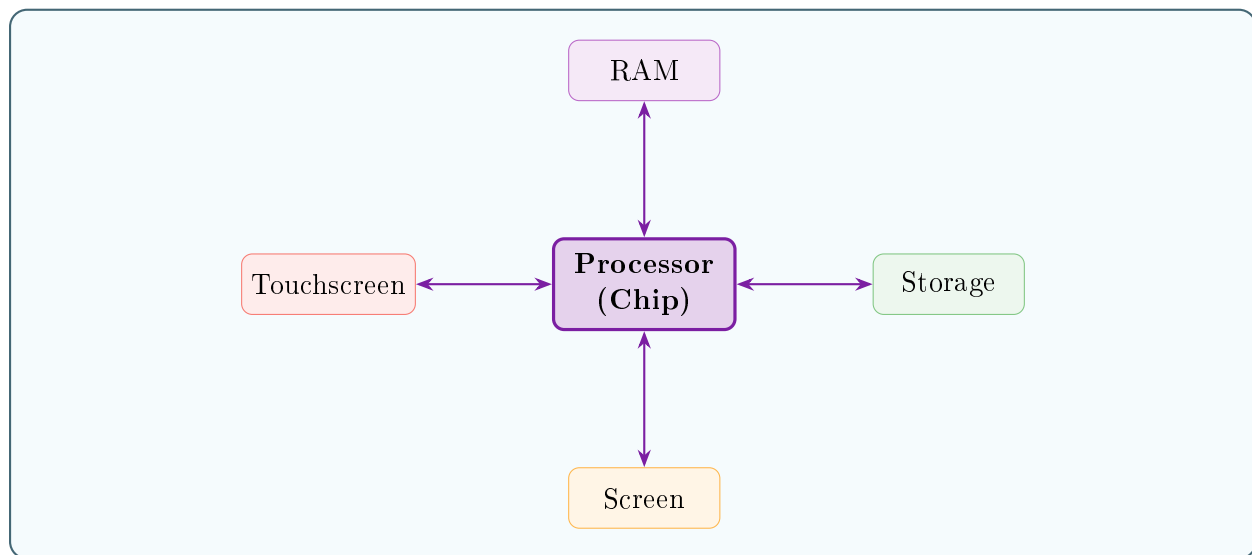
Billions of Tiny Switches

Inside a processor are billions of microscopic switches called **transistors**. Each transistor can be either **on** (1) or **off** (0). These ones and zeros are called **binary code** — the language all computers use. Groups of ones and zeros can represent numbers, words, colours, and sounds.



Did You Know?

A modern smartphone processor contains more than **10 billion transistors**. Each one is incredibly tiny. If you lined transistors up across a single human hair, thousands would fit.



The processor connects to all other parts of the phone.

Why Speed Matters

Processor speed is measured in **gigahertz** (GHz). One gigahertz means one billion steps per second.

Quick Map

What happens when you tap a photo to open it:

1. The touchscreen detects your tap and sends a signal.
2. The processor receives the signal (an instruction).
3. The processor fetches the photo data from storage.
4. The processor unpacks the image data into pixels.
5. The processor sends the image to the screen.
6. All of this happens in a fraction of a second.

New Words

Processor	The brain chip that carries out all instructions.
Transistor	A microscopic on/off switch inside a processor.
Binary code	The language of ones and zeros that computers use.
Gigahertz	A unit of speed: one billion steps per second.
Instruction	A single command the processor must carry out.

Try It

Be the processor!

Ask a friend to give you instructions one at a time (e.g., “Pick up the pencil”, “Move it to the table”, “Put it down”). Follow each instruction *exactly* as given. What happens if an instruction is unclear?

This is why programmers must write instructions very carefully.

Big Idea

The **processor** is the brain of the phone. It follows billions of instructions every second using tiny transistors and controls everything else inside the phone.

Chip

“I need somewhere to hold all the information I am working on. That is where my friend Memo comes in. On to Chapter 3!”

Chapter 3

Memory and Storage

Your phone remembers your photos, your apps, and your games. But did you know it uses two completely different kinds of memory to do it?

Mission: Spot the Difference Between RAM and Storage!

Meet Memo

Memo

“Hi! I am Memo — and I have two jobs. My first job is RAM: holding things Chip is working on right now. My second job is storage: keeping your photos and apps safe for as long as you need them. They sound similar, but they are very different!”

RAM: The Working Memory

RAM stands for **R**andom **A**ccess **M**emory. It is the phone’s *working memory* — the space where the processor keeps information it is actively using.



Imagine This

RAM is like a desk. When you are doing homework, you spread your books and papers across the desk. You can reach everything quickly. But when you finish and pack everything away, the desk is empty.

RAM is the same: it holds things while the phone is working on them, but when you switch the phone off, everything on the “desk” disappears.

When you open an app, it is loaded from storage into RAM so the processor can work with it quickly. The more apps you have open at once, the more RAM is being used — and the phone can slow down if RAM gets full. When RAM is very full, the phone may close an app running in the background to make space for the app you are using now.

Storage: The Long-Term Keep

Internal storage is where the phone keeps everything permanently: photos, videos, apps, messages, music, and the operating system itself.

Unlike RAM, storage *does not* get wiped when the phone is turned off. Your photos from last year are still there in the morning.



Imagine This

Storage is like a bookshelf or filing cabinet. You can put hundreds of books and folders there, and they stay exactly where you left them — even overnight, even for years. To read a book, you take it off the shelf (load it into RAM) — the shelf does not empty just because you picked something up.

Storage is measured in **gigabytes** (GB). Most smartphones have between 64 GB and 512 GB of storage.



Did You Know?

A single minute of 4K video can take up roughly 75 MB to 400 MB of storage, depending on quality settings. A 128 GB phone could hold about 5 to 28 hours of 4K footage — or around 32,000 songs, or 40 million text messages.

RAM vs Storage at a Glance

Quick Map

	RAM	Storage
Purpose	Working memory	Long-term keeping
Speed	Very fast	Slower than RAM
Cleared?	Yes, when phone is off	No, stays forever
Typical size	4–12 GB	64–512 GB
Analogy	A desk	A bookshelf

Why Phones Run Out of Space

Photos, videos, apps, and downloaded music all take up storage space. When storage gets full, the phone cannot save new things.

- Delete photos and videos you no longer need.
- Remove apps you rarely use.
- Back up files to a computer or the cloud (storage on internet-connected computers).



New Words

RAM

Random Access Memory — fast, temporary working space.

Storage

Permanent space for apps, photos, and files.

Gigabyte (GB)

A unit of data size: about 1 billion bytes.

Operating system

The main programme (iOS, Android) that manages the phone.



Try It

Find out how much storage your device has.

With a grown-up's help, go to the phone's *Settings* app and look for “Storage” or “About Phone”.

- How much total storage does it have?
- How much is already used?
- What is using the most space — photos, apps, or something else?



Big Idea

A smartphone uses two kinds of memory: **RAM** (fast, temporary working space cleared when off) and **storage** (permanent space for photos, apps, and files).

Memo

“Next, we will find out how you talk to the phone with your fingers. Meet Tappy in Chapter 4!”

Chapter 4

The Amazing Touchscreen

Every tap, swipe, and pinch you make on a phone screen is sending a signal to the processor. But how does a flat piece of glass know where your finger is?

Mission: Understand the Magic of Touch!

Meet Tappy

Tappy

“Hello! I am Tappy — and I live right here on the surface of the screen. Every time a finger lands on the glass, I feel it. I know exactly where it touched, how many fingers touched at once, and how long they stayed. Then I tell Chip what to do next.”

How Touching the Screen Sends Commands

A smartphone screen is a **capacitive touchscreen**. “Capacitive” means it works with an invisible electrical field spread across the glass.

Your skin can conduct a tiny amount of electricity. When your finger touches the screen:

1. The screen’s invisible grid of sensors detects a change in that electrical field.
2. The sensors record the exact location of the touch.
3. That information is sent to the processor as an instruction.
4. The processor decides what to do (open an app, scroll the page, etc.).

Imagine This

Imagine the screen is covered by an invisible net of tiny tripwires. When your finger lands, it triggers the tripwires nearby. The phone measures *which* wires were triggered to figure out exactly where you touched.

★ Did You Know?

Touchscreens do **not** work with gloves on (usually) — because the glove blocks the electrical charge from your finger. But special “touchscreen gloves” are made with conductive material (material that lets electricity pass through) in the fingertips so the signal passes through.

Swiping, Tapping, and Pinching

Different finger movements (called **gestures**) send different instructions.

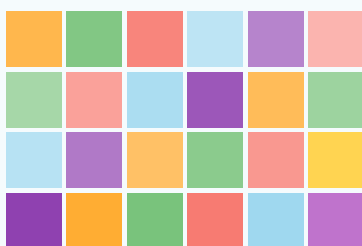
Quick Map

Gesture	What it usually does
Tap	Select or open something
Double tap	Zoom in or like a photo
Swipe	Scroll or switch between screens
Pinch in	Zoom out
Spread out	Zoom in
Long press	Open extra options
Drag	Move something across the screen

The Screen Itself: Millions of Tiny Lights

Beneath the touch sensors is the actual display. It is made up of millions of tiny dots of light called **pixels**. Each pixel can be set to any colour. When millions of pixels light up together, they form the images you see.

On the same-sized screen, more pixels means a sharper, more detailed image. The number of pixels on a screen is its **screen resolution**.



Each small square is one pixel. Together they form a picture.

A grid of pixels makes up every image on the screen.



New Words

Capacitive touchscreen	A screen that detects where your finger changes its invisible electrical field.
Gesture	A finger movement that sends a command.
Pixel	A tiny dot of colour on a screen.
Screen resolution	The number of pixels on a screen.



Experiment

Glove test!

Try tapping the screen of a smartphone with:

1. Your bare finger.
2. A rubber glove or washing-up glove.
3. A knitted glove.
4. The eraser end of a pencil.

Which ones work? Which ones do not? Can you explain why?

Hint: think about whether each material conducts electricity.



Big Idea

A touchscreen detects the **electrical charge** from your fingertip. Different finger gestures send different commands to the processor, which decides what happens on screen.

Tappy

“You are a great communicator! Now let’s visit another amazing part of the phone — the cameras. Meet Lens in Chapter 5!”

Chapter 5

Tiny Cameras, Big Pictures

One small hole in the back of your phone can capture a moment forever. How does such a tiny camera take such sharp, colourful photos?

Mission: Capture the World with Light!

Meet Lens

Lens

“Hello! I am Lens — and light is my favourite thing. I bend it, focus it, and send it exactly where it needs to go. Without me, every photo would be a blurry mess. Together with the sensor behind me, we turn a moment into a memory.”

Lenses and Sensors

Every smartphone camera has two main parts working together.

The Lens

A **lens** is a curved piece of glass or plastic. When light enters the camera, the lens bends it so that it focuses into a sharp, clear image on the sensor behind it. The lens helps light from each tiny part of the scene land in the right place, so details do not smear together.

Without a lens, the light would scatter and produce a blurry image.

The Camera Sensor

Behind the lens sits the **camera sensor**. It is covered in millions of tiny light-detecting spots called **pixels**. When light hits the sensor, each pixel records how bright the light is and what colour it is. All those tiny measurements combine to form a complete photograph. Then the processor quickly combines all that information into the photo you see.



Imagine This

Imagine holding a piece of graph paper up to a colourful painting. Each square on the grid captures one tiny patch of colour. The more squares there are, the more detail you get. A camera sensor works the same way — each pixel records one tiny patch of the scene.



Did You Know?

A modern smartphone camera sensor can have **50 megapixels** — that is 50 *million* tiny light detectors, all in a chip smaller than your thumbnail.

How Selfies Work

Most smartphones have **two cameras**:

- The **rear camera** (on the back) — usually higher quality, used for general photos.
- The **front camera** (on the screen side) — used for selfies and video calls.

When you tap the camera-flip button, the phone simply switches which sensor is active. The front camera works in exactly the same way as the rear one.

Night Mode and Zoom

Night Mode

In low light, a camera sensor does not receive much light per pixel, making photos look dark or grainy.

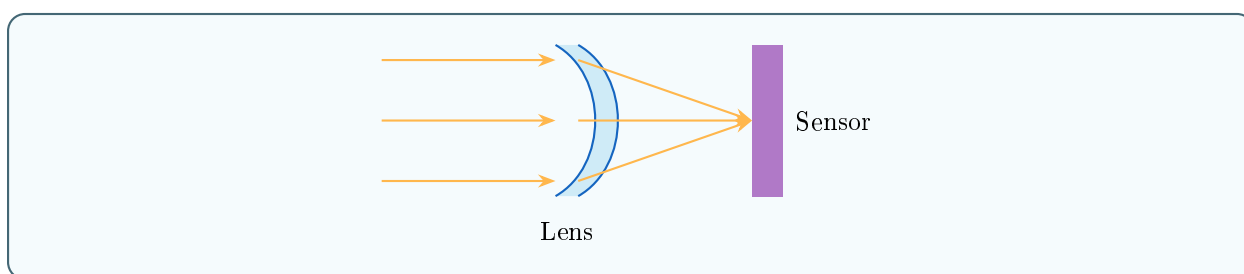
Night mode solves this by taking many photos very quickly and combining them into one image. The processor averages out the light information across all the shots, revealing details invisible in any single photo.

Zoom

There are two types of zoom:

- **Optical zoom:** the lens physically moves to magnify the image. This keeps the photo sharp.
- **Digital zoom:** the processor crops and enlarges the image. This can make the photo look less sharp.

Many smartphones have two or three cameras with different focal lengths, so they can switch lenses to get true optical zoom.



How light travels through a lens to the sensor.



New Words

Lens	A curved piece of glass that focuses light.
Camera sensor	The chip that converts light into digital information.
Pixel	One tiny light-detecting spot on a sensor or screen.
Night mode	A feature that combines multiple shots for brighter low-light photos.
Optical zoom	Zoom achieved by moving the lens, keeping images sharp.
Digital zoom	Zoom achieved by cropping and enlarging the image.
Selfie	A photo taken of yourself with the front-facing camera.



Experiment

Light and shadow!

Take a photo of the same object in three conditions:

1. In bright daylight or near a window.
2. In a dimly lit room.
3. In a dimly lit room with night mode turned on.

Compare the three photos. What differences do you notice in sharpness, brightness, and colour? Can you explain why?



Big Idea

A smartphone camera uses a **lens** to focus light onto a tiny **sensor** packed with millions of pixels. Smart software like night mode helps the processor produce clear photos even in difficult conditions.

Lens

“A great shot needs power too. Let’s visit Zap and learn about the battery in Chapter 6!”

Chapter 6

The Battery Power Station

Every time you charge your phone, you are filling up a tiny power station. Where does that energy go, and why doesn't it last forever?

Mission: Power Up and Understand Energy!

Meet Zap

Zap

“ZAP! That’s me — your battery. Without me, none of the others could do a thing. Chip would go dark. Lens would see nothing. I store the energy the whole phone needs, and I send it where it is needed. But I work hard, and I need recharging. Let me show you how!”

Without a battery, a smartphone is just a quiet piece of glass and metal. The battery is what brings everything to life.

Where Energy Comes From

A smartphone battery is a **lithium-ion battery**. It does not store electricity directly — it stores **chemical energy**.

Inside the battery, a chemical reaction moves tiny charged particles (called electrons) from one side to the other. As they flow, they create an **electric current** — the electricity that powers the phone.



Imagine This

Imagine a water tank on a hill. Water flows downhill through a pipe, turning a wheel to produce power. The battery is the tank: it stores potential energy. When the phone needs power, the energy “flows” out. When you plug in the charger, you are pumping the tank back up.

Charging Your Phone

When you plug in a charger, electricity from the wall socket reverses the chemical reaction inside the battery. This refills the battery’s stored energy.

Quick Map

From wall socket to a charged phone:

1. Electricity travels from the power socket through the charger cable.
2. The charger converts the voltage to a safe level for the battery.
3. The electric current reverses the chemical reaction inside the battery.
4. The battery fills up with stored chemical energy.
5. When full, the charger stops sending current automatically.



Did You Know?

Some modern smartphones can charge to 50% in just **15 minutes** using fast-charging technology. Compare that to the first mobile phone batteries, which took several hours.

Why Batteries Don't Last Forever

Each time a lithium-ion battery charges and discharges, the chemical materials inside change very slightly. Over hundreds of charge cycles, the battery holds a little less energy than before.

After 2–3 years of daily use, a battery might only hold 80% of its original charge. Eventually it needs to be replaced.

Saving Battery Life

Some things drain the battery faster than others:

- Bright screen at full brightness
- Streaming video
- GPS navigation (you will meet GPS in Chapter 9)
- Running many apps at once

Ways to save battery:

- Lower screen brightness.
 - Turn on battery-saving mode.
 - Close apps you are not using.
 - Turn off Wi-Fi, Bluetooth, and GPS when not needed.
-



New Words

Lithium-ion battery	The type of rechargeable battery used in smartphones.
Chemical energy	Energy stored in the form of a chemical reaction.
Electric current	The flow of electrons that powers the phone.
Charging	Restoring energy to a battery using an external power source.



Experiment

Battery drain test!

Start with a fully charged phone. Check and record the battery level every hour.

- Hour 1: Screen off, doing nothing.
- Hour 2: Playing a video game.
- Hour 3: Watching a video with high brightness.
- Hour 4: In battery-saving mode.

Which activity drained the battery fastest? Which was most efficient? Why?

Big Idea

A smartphone **battery** stores chemical energy and converts it into electricity. Charging reverses this reaction to refill the energy. Batteries wear out slowly over time because of repeated charge cycles.

Zap

“Powering up is my job. But the phone also needs to sense the world. Meet Sensa in Chapter 7!”

Chapter 7

The Hidden Sensors

Your phone can feel you turning it sideways. It dims its screen in your pocket. It counts your steps without you pressing anything. Meet the invisible helpers doing all of this — the sensors.

Mission: Feel the World Through Your Phone!

Meet Sensa

Sensa

“I am Sensa — actually, I am a whole family of tiny sensors! Most people never notice us, but we are always watching and listening. We tell the phone which way is up, how fast it is moving, how bright the room is, and which direction is north. Without us, your phone would be blind to the physical world.”

Sensor 1: The Accelerometer

The **accelerometer** measures how fast the phone is speeding up or slowing down — and in which direction. Because gravity always pulls downward, it also helps the phone work out which way is up.

- It detects when you rotate the phone from portrait to landscape and tells the screen to rotate too.
- It counts your steps in a fitness app by detecting each footfall.
- It detects a sudden drop and can take protective action.



Experiment

Hold a phone and slowly rotate it from portrait (upright) to landscape (sideways). Watch the screen rotate. That is the accelerometer at work!

Sensor 2: The Gyroscope

The **gyroscope** measures how the phone is *rotating* — not just tilting, but spinning around any axis. The accelerometer notices movement, while the gyroscope measures rotation in fine detail.

- Games use the gyroscope so you can steer by tilting the phone.
- Panorama photo apps use it to stitch a wide photo accurately.
- Virtual reality apps use it to track which direction you are looking.



Imagine This

Imagine spinning a toy top on a table. A gyroscope inside the phone works like a tiny, invisible top — it resists changes in direction, letting the phone measure exactly how it has turned.

Sensor 3: The Light Sensor

The **light sensor** (also called the ambient light sensor) measures how bright the surroundings are.

- On a sunny day, it tells the screen to get brighter so you can read it.
- In a dark room, it dims the screen to save your eyes and battery.
- It helps automatic brightness keep the screen comfortable to look at.

Phones also have a **proximity sensor**, which helps turn the screen off when the phone is close to your face during a call.

Sensor 4: The Microphone

The **microphone** is a sensor that converts sound waves into electrical signals.

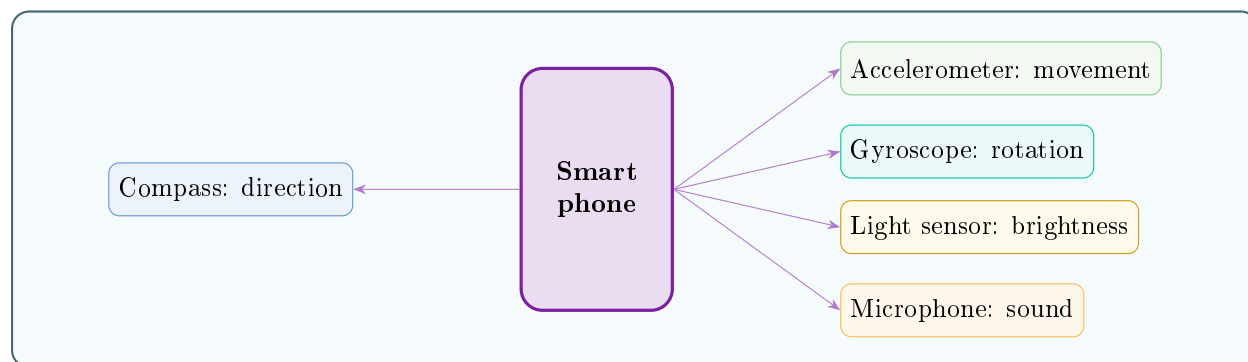
- It lets you make calls and record voice notes.
- It powers voice assistants that listen for your commands.
- It records audio during video recording.

Most smartphones have two or three microphones to capture sound from different directions and reduce background noise.

Sensor 5: The Compass

The **compass** sensor (called a magnetometer) detects the direction of Earth’s magnetic field — just like a traditional compass needle.

- It tells your maps app which direction you are facing.
- It helps GPS navigation show the correct arrow direction.
- Augmented reality apps use it to overlay digital objects on the real world.



The five hidden sensors and what each one detects.

New Words

Accelerometer	Sensor that measures speed and direction of movement.
Gyroscope	Sensor that measures rotation.
Light sensor	Sensor that measures ambient brightness.
Microphone	Sensor that converts sound waves into electrical signals.
Compass	Sensor that detects Earth’s magnetic field to find direction.
Sensor	A device that detects changes in the physical world.

Big Idea

A smartphone has many **hidden sensors** that detect the physical world: movement, rotation, light, sound, and magnetic direction. They work quietly in the background to make the phone smarter.

Sensa

“We sensors feed information to Chip so the phone can react to the world. Next, let’s meet the programmes that put all that power to work. Meet Appy in Chapter 8!”

Chapter 8

Apps: The Workers Inside Your Phone

The hardware — the processor, screen, and sensors — is like an empty factory. Apps are the workers who give that factory a purpose.

Mission: Put the Workers to Work!

Meet Appy

Appy

“Hi! I am Appy, and I keep busy every second of every day. I am a programme — a set of instructions that tells Chip exactly what to do. Without me, your phone’s hardware would just sit there doing nothing. Every game, map, music player, and camera programme — that is an app like me!”

What Apps Do

An **app** (short for *application*) is a programme designed to do a specific job on a smartphone. Apps are written by developers — people who write instructions in code.

Apps use the phone's hardware to do their work:

- A maps app uses the GPS sensor, screen, and internet connection.
- A camera app uses the camera sensor, touchscreen, and storage.
- A music app uses the storage, processor, and speaker.



Imagine This

Think of the phone's hardware as a well-equipped kitchen: it has an oven, fridge, knives, and pans. An app is the chef who decides what to cook and how to use every tool. Without the chef, the kitchen does nothing. Without the kitchen, the chef has nothing to work with.

The Operating System: The Manager

Apps do not work alone. They all depend on the **operating system** (OS).

The operating system (like Android or iOS) is the main programme that:

- Manages all the hardware (screen, processor, battery).
- Lets multiple apps run without getting in each other's way.
- Controls which apps are allowed to use which sensors or data.
- Provides the home screen, notifications, and settings.

Apps are built on top of the operating system — they cannot bypass it. That keeps the phone organised and helps protect your data.

Apps for Every Purpose

Quick Map

Apps by category:

Category	Examples
Communication	Messaging, email, video calls
Entertainment	Games, music, video streaming
Learning	Educational, language, maths
Navigation	Maps, GPS directions
Creativity	Drawing, photo editing, music making
Productivity	Notes, calendars, to-do lists
Health	Step counter, sleep tracker

Installing and Updating Apps

Apps are downloaded from an **app store** — a digital shop where developers post their apps for people to install. Some apps are free to download but make money through adverts or extra paid features. That is why it is smart to ask a trusted adult before buying anything in an app.

A **update** is a new version of an app that:

- Fixes bugs (small errors in the code).
- Adds new features.
- Improves security.

Keeping apps updated is important — old versions can have security weaknesses.

★ Did You Know?

As of 2024, there are more than **3.5 million apps** available on the Google Play Store alone. Developers around the world upload thousands of new apps every single day.

📖 New Words

App	A programme that runs on a phone to do a specific job.
Operating system	The main programme managing all hardware and apps (iOS, Android).
Update	A new version of an app or OS that fixes bugs and adds features.
App store	A digital shop where apps are downloaded and installed.

Try It

App audit!

With a grown-up's help, look at the apps installed on a smartphone.

1. Count how many apps are installed.
2. Group them into categories (games, learning, communication, etc.).
3. Which category has the most apps?
4. Are there any apps that have never been opened? Should they be deleted?

Big Idea

An **app** is a programme that tells the phone's hardware what to do. The **operating system** manages all apps and hardware. Keeping apps **updated** keeps the phone working well and safely.

Appy

"I can help you find your way, but only because of one very special helper high above the Earth. Meet Orbi in Chapter 9!"

Chapter 9

GPS: Finding Your Way

You can be in the middle of a forest with no Wi-Fi and your phone still knows exactly where you are. How is that possible?

Mission: Navigate by Satellite!

Meet Orbi

Orbi

“Greetings from orbit! I am Orbi — one of dozens of GPS satellites circling the Earth right now, thousands of kilometres above you. I send a very precise signal to your phone every second. Your phone listens to me and a few of my friends, and together we tell it exactly where it is on Earth.”

What Is GPS?

GPS stands for **Global Positioning System**. It is a network of more than 30 satellites orbiting Earth. They were originally built by the United States military, but today anyone with a GPS receiver — including every smartphone — can use them for free.

★ Did You Know?

GPS satellites orbit about **20,200 kilometres** above Earth. They travel so fast that they orbit the entire planet twice a day. Each satellite sends a signal that travels at the speed of light.

How Phones Know Where They Are

Your phone has a tiny GPS **receiver** that listens for signals from satellites. Each satellite constantly broadcasts two pieces of information:

- Its exact location in space.
- The exact time the signal was sent.

The phone receives these signals and measures how long each signal took to arrive. Since signals travel at the speed of light, even a tiny delay tells the phone how far away each satellite is. The phone uses those distances to calculate its position.

Triangulation: Pinpointing Your Location

The process of using multiple satellites to find a position is called **triangulation**.



Imagine This

Imagine you are lost in a park and you can hear three different ice cream vans. You can tell which direction each van is and roughly how far away it sounds. Using all three distances, you can work out exactly where you must be standing. GPS works the same way — but with satellites instead of ice cream vans, and light-speed radio signals instead of music.

Quick Map

How triangulation finds your location:

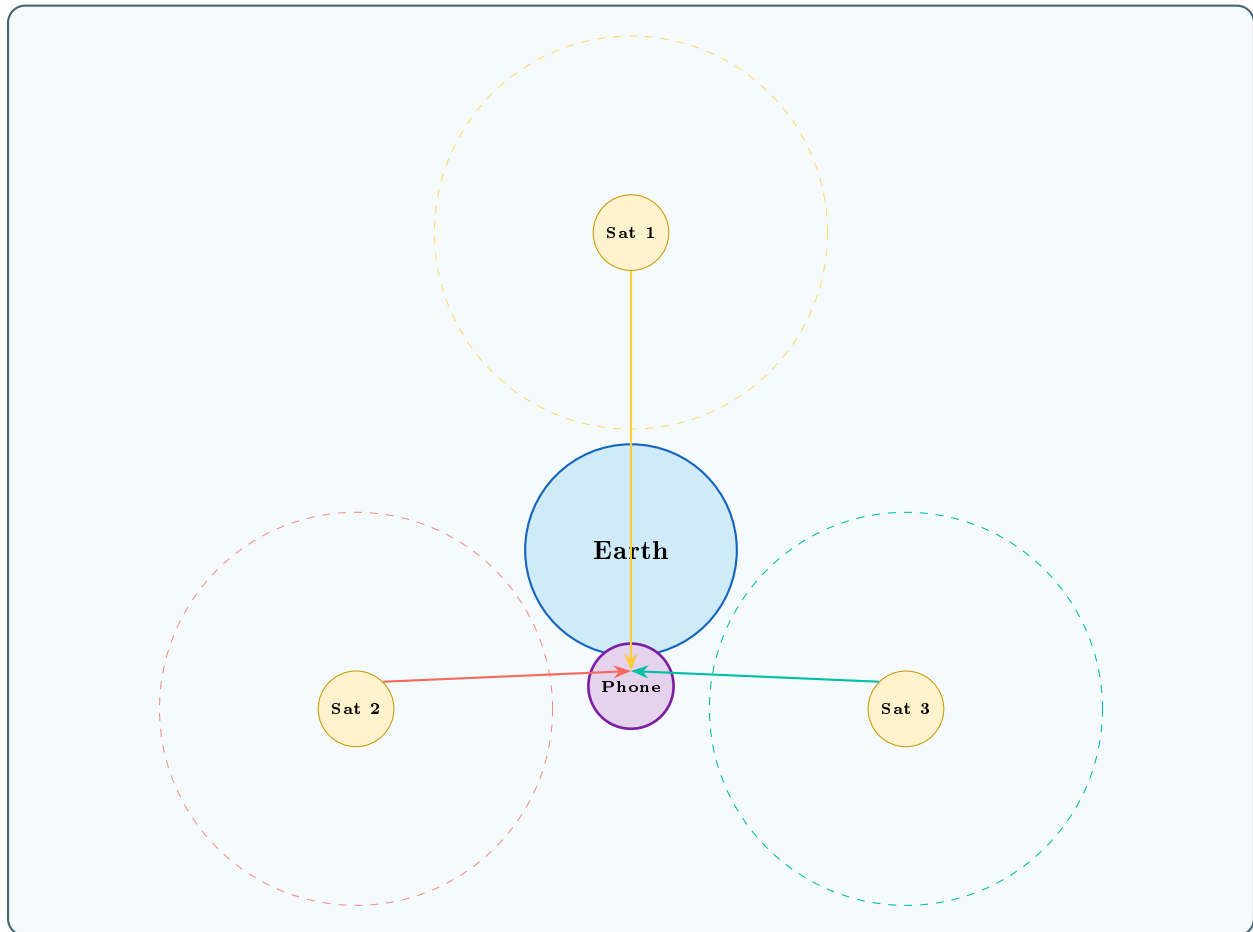
1. Satellite A sends a signal. The phone knows it is *some distance* from A.
2. Satellite B sends a signal. The phone narrows the position down further.
3. Satellite C sends a signal. The position is now pinpointed very precisely.
4. A fourth satellite confirms the exact altitude (height above sea level).

GPS gives your phone location numbers (latitude and longitude), then your maps app turns those numbers into the map view you recognise.



New Words

GPS	Global Positioning System — a satellite network for finding location.
Satellite	A machine orbiting Earth that sends or receives signals.
Receiver	A device that picks up GPS signals (built into every smartphone).
Triangulation	Using signals from three or more sources to find an exact position.



Three satellites triangulate a phone's position.

Try It

Find yourself on the map!

With a grown-up's help, open a maps app on a smartphone and allow it to use your location.

- Walk to a different room. Does the dot move?
- Go outside and walk half a block. How accurate is the position?
- Notice the small circle around the dot — that shows where the phone thinks you might be.

Can you figure out which direction is north by moving in a straight line?

Big Idea

GPS uses signals from multiple satellites orbiting Earth to calculate a phone's exact position through **triangulation**. It works anywhere on Earth, with no Wi-Fi or mobile signal needed.

Orbi

“You can now find your way anywhere on the planet! But with great power comes great responsibility. Our final chapter is about staying safe and smart. Follow Dot!”

Chapter 10

Stay Safe and Smart

You have explored every part of a smartphone. Now it is time to learn how to use this powerful tool wisely, safely, and kindly.

Final Mission: Become a Smart Smartphone Explorer!

The Smartphone Explorer's Code

All of the characters gathered around Sam one last time: Chip, Memo, Tappy, Lens, Zap, Sensa, Appy, Orbi — and of course, Dot.

Dot

“You have come a long way since that first question: “How does all of this fit inside one tiny phone?” Now you know what is inside the tiny computer in your pocket. But knowing how something works also means knowing how to use it responsibly. Are you ready for your final challenge?”

Tool 1: A Strong Password

A **password** protects your accounts from people who should not access them. A strong password:

- Is long (at least 12 characters).
- Mixes letters, numbers, and symbols.
- Is different for every account.
- Is never shared with anyone except a trusted adult.



Imagine This

A weak password is like a front door left wide open. A strong password is a sturdy, locked door. And sharing your password with someone you do not trust is like giving a stranger the key to your home.

Tool 2: Fingerprints and Face Unlock

Biometrics uses a unique part of your body to identify you. Smartphones can unlock using:

- **Fingerprint:** A sensor scans the unique ridges on your fingertip.
- **Face unlock:** Cameras and sensors map the unique shape of your face.

Biometrics are convenient because you always have them with you — but a password is still a good backup. If biometrics ever fail, your password helps you get back into your device safely.

Tool 3: Asking Permission Before Downloading Apps

When you install an app, it may ask for **permissions** — access to parts of your phone like the camera, microphone, or location.

- Always read what an app is asking permission for.
- Ask yourself: does this app really need this access? (A torch app does not need your location.)
- Ask a trusted adult before installing unfamiliar apps.
- Only download from trusted app stores.

Quick Map

Smart permission choices:

Situation	Smart?
Maps app asks for your location.	Yes!
A game asks to read all your contacts.	Ask an adult.
A torch app asks to use your microphone.	Suspicious!
You update your apps regularly.	Great idea!

Tool 4: Being Kind Online

A smartphone connects you to real people all over the world.

- Treat people online the same way you would face to face.
- Think before you send: “*Would I say this in person?*”
- If someone is unkind to you online, it is not your fault. Tell a trusted adult.

Appy

“Messages sent through me reach real people with real feelings. A kind message can make someone’s day. Think before you type.”

Tool 5: Balance Your Screen Time

Smartphones are powerful tools — but so is the world away from screens.

- Take regular breaks from your phone.
- Notice how you feel after a long session.
- Balance phone time with outdoor play, reading, and time with family and friends.

- Ask yourself: “Am I choosing to use my phone right now, or just using it out of habit?”
-

Smartphone Explorer’s Code of Honour

Imagine This

A great explorer does not just learn facts — they make promises. Read this code and sign it as a true smartphone explorer.

1. I will protect my passwords and keep personal information private.
2. I will check app permissions carefully before installing anything new.
3. I will keep my apps and operating system updated.
4. I will be kind in every message, comment, and online game.
5. I will ask a trusted adult for help if something feels wrong.
6. I will balance screen time with rest, play, and real-world adventures.

Explorer’s Signature: _____

Date: _____

You Are a Smartphone Explorer!

All the characters gathered around Sam and cheered.

Dot

“You started this journey wondering what was inside a tiny phone. Now you know about the processor, memory, touchscreen, cameras, battery, sensors, apps, and GPS. You understand how they work together — and how to use them wisely. Take that knowledge and go explore the world!”

Big Idea

A **smart smartphone user** understands their device, protects their information, treats others with kindness, and balances screen time with the rest of life.

Big Ideas to Remember

1. Smartphones are tiny but powerful computers.

2. The processor is the brain that follows instructions.

3. RAM is working memory; storage keeps things safe long-term.

4. Touchscreens sense the tiny electrical charge in your fingertip.

5. Cameras use lenses and sensors to capture light.

6. Batteries store chemical energy and convert it to electricity.

7. Hidden sensors help phones understand movement and the world.

8. Apps are programmes that run inside the phone to do specific jobs.

9. GPS satellites help your phone know exactly where you are.

10. Smart users protect their phones, their data, and their wellbeing.

Glossary



All the Words You Learned

Accelerometer	A sensor that measures how fast a phone is speeding up or slowing down. (<i>Chapter 7</i>)
App	A programme that runs on a smartphone to do a specific job — like playing music or showing a map. (<i>Chapter 8</i>)
Battery	A device that stores chemical energy and converts it into electricity to power the phone. (<i>Chapter 6</i>)
Biometrics	Using a unique part of your body — like your fingerprint or face — to unlock a device. (<i>Chapter 10</i>)
Camera sensor	The part of a camera that converts light into digital information. (<i>Chapter 5</i>)
Capacitive touchscreen	A type of screen that detects the tiny electrical charge in a human fingertip. (<i>Chapter 4</i>)
Charging	Restoring energy to a battery by connecting it to a power source. (<i>Chapter 6</i>)
Clock speed	How many instructions a processor can carry out every second, measured in gigahertz (GHz). (<i>Chapter 2</i>)
Compass	A sensor that detects Earth's magnetic field to determine which direction the phone is facing. (<i>Chapter 7</i>)
GPS	Global Positioning System — a network of satellites that helps devices find their location. (<i>Chapter 9</i>)
Gyroscope	A sensor that measures the rotation and tilt of a phone. (<i>Chapter 7</i>)
Internal storage	The permanent memory inside a phone that keeps photos, apps, and files even when the phone is off. (<i>Chapter 3</i>)
Lens	A curved piece of glass or plastic that focuses light onto the camera sensor. (<i>Chapter 5</i>)
Light sensor	A sensor that measures the brightness of the surroundings and adjusts screen brightness automatically. (<i>Chapter 7</i>)

Lithium-ion battery	The common type of rechargeable battery used in smartphones. <i>(Chapter 6)</i>
Microphone	A sensor that converts sound waves into electrical signals so the phone can record or transmit voice. <i>(Chapter 7)</i>
Night mode	A camera feature that takes multiple photos quickly and combines them to produce a brighter image in low light. <i>(Chapter 5)</i>
Operating system	The main programme that manages all the hardware and software on a phone (e.g., Android, iOS). <i>(Chapter 8)</i>
Pixel	The smallest dot of colour on a screen or in a digital image. More pixels means more detail. <i>(Chapter 5)</i>
Processor	The brain of the phone — a tiny chip that carries out instructions and controls everything the phone does. <i>(Chapter 2)</i>
RAM	Random Access Memory — the phone’s working memory, used to hold information the phone is actively using right now. <i>(Chapter 3)</i>
Satellite	A machine that orbits Earth and sends or receives signals to and from devices on the ground. <i>(Chapter 9)</i>
Screen resolution	The number of pixels on a screen. Higher resolution means sharper images. <i>(Chapter 4)</i>
Selfie	A photo taken of yourself, usually with the front-facing camera. <i>(Chapter 5)</i>
Sensor	A tiny device inside a phone that detects changes in the physical world — like movement, light, or direction. <i>(Chapter 7)</i>
Smartphone	A mobile phone that is also a computer — it can run apps, connect to the internet, and use cameras and sensors. <i>(Chapter 1)</i>
Storage	The space used to save information permanently on a phone. <i>(Chapter 3)</i>
Touchscreen	A screen that can detect when and where you touch it, turning your finger into a control. <i>(Chapter 4)</i>
Transistor	A microscopic switch inside a processor that can be on or off — billions of them work together to process information. <i>(Chapter 2)</i>
Triangulation	Using signals from three or more satellites to calculate an exact position on Earth. <i>(Chapter 9)</i>
Update	A new version of an app or the operating system that fixes problems or adds features. <i>(Chapter 8)</i>
Zoom	Making part of an image appear larger. Digital zoom crops the

image; optical zoom uses moving lenses. (*Chapter 5*)

Quiz Time!

You have explored every corner of a smartphone — from the processor to the GPS satellites far above! Now test what you have learned.

Part 1: True or False?

Write **T** (true) or **F** (false) next to each statement.

1. A smartphone is just a phone — it cannot do anything a computer can do. _____
2. The processor carries out instructions and controls the whole phone. _____
3. RAM keeps your photos saved even when the phone is switched off. _____
4. Touchscreens detect the heat from your finger. _____
5. A camera sensor converts light into digital information. _____
6. Batteries store chemical energy and convert it into electricity. _____
7. The gyroscope tells your phone which direction is north. _____
8. GPS works by receiving signals from satellites orbiting Earth. _____
9. You need at least three satellites to find your exact position. _____
10. Using the same password for every app is a good idea. _____

Part 2: Match the Character to Their Job

Draw a line from each character to what they do inside the phone:

Character

1. Dot
2. Chip
3. Memo
4. Tappy
5. Lens
6. Zap
7. Sensa
8. Appy
9. Orbi

Job

- A. Runs your favourite apps and games
- B. A returning guide — a packet of information
- C. Captures photos and videos
- D. The brain that follows all instructions
- E. Stores photos and apps permanently
- F. Detects the electrical charge in your fingertip
- G. Powers the phone with stored energy
- H. Feels every tilt, twist, and direction change
- I. Orbits Earth and helps the phone find its location

Part 3: Fill in the Blanks

Use the word bank to complete each sentence.

RAM processor accelerometer satellite pixels operating system

1. The _____ is the brain of the phone that carries out billions of instructions every second.
2. _____ is the working memory that holds information the phone is actively using.
3. A photo is made up of millions of tiny dots of colour called _____.
4. GPS works by receiving signals from at least three _____s orbiting Earth.
5. The _____ measures how fast your phone is speeding up or slowing down.
6. The _____ is the main programme (like Android or iOS) that manages the whole phone.

Part 4: Put It in Order

Number these steps 1 to 6 to show what happens when you take a photo and save it:

- _____ The processor compresses the image file.
- _____ You tap the shutter button on the screen.
- _____ The image is saved to internal storage.
- _____ Light passes through the lens onto the camera sensor.
- _____ The camera sensor converts light into digital information.
- _____ The touchscreen detects your finger and sends the command.

Part 5: Smartphone Word Challenge

Smartphone Word Challenge

Across clues:

1. The brain of the phone that follows instructions (9 letters)
2. A network of orbiting machines used to find your location (3 letters)
3. The working memory that holds what the phone is doing right now (3 letters)
4. A programme that runs on your phone to do a specific job (3 letters)
5. A tiny dot of colour in a photo or on a screen (5 letters)

Down clues:

1. The permanent memory where photos and apps are saved (7 letters)
2. A sensor that measures how the phone is tilting or rotating (9 letters)
3. The part of the camera that focuses light (4 letters)
4. A device that orbits Earth and sends GPS signals (9 letters)
5. A device that stores energy and powers the phone (7 letters)

Part 6: Think About It

Answer these questions in your own words:

1. Why does a phone slow down when you open lots of apps at the same time?
2. GPS works even when there is no Wi-Fi. How is that possible?
3. Why do smartphone cameras take better photos in daylight than at night?
4. You download a new game and your phone says “Storage full.” What could you do?
5. What are two ways to keep your smartphone and your personal information safe?

Answer Key

Answers (no peeking until you try!)

Part 1: 1. F 2. T 3. F 4. F 5. T 6. T 7. F 8. T 9. T 10. F

Part 2: 1-B 2-D 3-E 4-F 5-C 6-G 7-H 8-A 9-I

Part 3: 1. processor 2. RAM 3. pixels 4. satellite 5. accelerometer 6. operating system

Part 4: 3, 1, 6, 2, 4, 5

Part 5 Across: 1. PROCESSOR 2. GPS 3. RAM 4. APP 5. PIXEL

Part 5 Down: 1. STORAGE 2. GYROSCOPE 3. LENS 4. SATELLITE
5. BATTERY

Parent & Teacher Guide

How to use this guide: Each chapter entry includes discussion questions to explore together, extension activities for deeper learning, and notes on curriculum areas supported. Read one chapter at a time and allow space for conversation between chapters. This book builds on concepts from *The Secret Life of the Internet* (Book 1) but can also be read as a standalone.

Chapter 1: What Is a Smartphone?

Discussion Questions

1. Name five different things you can do with a smartphone. Would you have needed five different devices 30 years ago?
2. How do you think life is different for people who do not have a smartphone?
3. A smartphone is described as a tiny computer. What makes it similar to a laptop? What makes it different?

Extension Activities

- **Feature audit:** List every feature and sensor your household phone has and categorise them (communication, navigation, camera, etc.).
- **Timeline:** Research the history of mobile phones from the 1970s brick phone to today's smartphones. Create a visual timeline.
- **Comparison chart:** Compare a smartphone to a 1990s desktop computer — processing speed, storage, camera, portability.

Curriculum Links

Science & Technology, History (technological change), PSHE (digital citizenship).

Chapter 2: Meet the Brain: The Processor

Discussion Questions

1. Why do you think processor speed is measured in *billions* of instructions per second? Can you imagine what one billion of anything looks like?
2. Why do some apps make your phone warm? Where does that heat come from?
3. What would happen if a phone had a very slow processor?

Extension Activities

- **Instruction game:** Give a child a very detailed set of instructions for a simple task (e.g., making a sandwich). Time them. Discuss how computers must follow instructions even more precisely.
- **Research:** Find out how many transistors are in a modern smartphone processor versus a 1970s CPU.
- **Human CPU:** Act out a processor — one person gives instructions, another executes them, a third records the result.

Curriculum Links

Mathematics (large numbers, scale), Computing (algorithms, logical thinking), Science (heat energy).

Chapter 3: Memory and Storage

Discussion Questions

1. What is the difference between RAM and storage in your own words?
2. Why do you think photos and videos take up more storage than text messages?
3. What would happen if you deleted all your apps — would the phone still work?

Extension Activities

- **Try It:** Check the storage on a device. How much is used? What is using the most space?
- **Comparison:** Compare file sizes — a text document, a photo, a song, and a video. Why are they so different?
- **Analogy building:** Ask children to invent their own analogy for RAM vs storage (not a desk — something from their own life).

Curriculum Links

Mathematics (units: KB, MB, GB, TB), Computing (data representation), Science (capacity).

Chapter 4: The Amazing Touchscreen

Discussion Questions

1. Why does a touchscreen not work when you use a gloved hand?
2. Why do some apps respond differently to a swipe versus a tap?
3. Can you think of other surfaces in everyday life that respond to touch?

Extension Activities

- **Glove test:** Try using a touchscreen with a rubber glove, then a knitted glove, then a bare finger. Discuss why each behaves differently.
- **Gesture dictionary:** List and define all the touch gestures you know (tap, swipe, pinch, spread, long press, etc.).
- **Design challenge:** Invent a new touchscreen gesture and describe what it should do in an app.

Curriculum Links

Science (electricity, conductivity), Computing (human-computer interaction), Art (design thinking).

Chapter 5: Tiny Cameras, Big Pictures

Discussion Questions

1. Why do phone cameras keep improving even though the phone size stays the same?
2. What is the difference between digital zoom and optical zoom? Which produces a better image?
3. How do you think night mode “finds” more light when there is not much there?

Extension Activities

- **Light experiment:** Photograph the same scene in bright light, dim light, and with night mode on. Compare the results.
- **Pixel art:** Create pixel art on squared paper to understand how images are made of tiny coloured squares.
- **Pinhole camera:** Build a simple pinhole camera from a cardboard box to see how lenses focus light.

Curriculum Links

Science (light, optics), Art (photography, composition), Computing (digital images, data).

Chapter 6: The Battery Power Station

Discussion Questions

1. Why do batteries wear out over time and need replacing after a few years?
2. Which things use the most battery on a phone? How could you test this?
3. Where does the electricity that charges your phone originally come from?

Extension Activities

- **Battery diary:** Track battery percentage at set times throughout the day. Which activities drain it fastest?
- **Lemon battery:** Build a simple lemon battery to demonstrate that chemical reactions can produce electricity.
- **Energy chain:** Trace the energy chain from a power station to the phone screen.

Curriculum Links

Science (energy transfer, chemical energy, electricity), Geography (energy sources), PSHE (environmental impact).

Chapter 7: The Hidden Sensors

Discussion Questions

1. Why does your screen turn off automatically when you hold the phone to your ear? Which sensor does that?
2. How do fitness apps know how many steps you have taken without you pressing anything?
3. If you were designing a new sensor for smartphones, what would it detect?

Extension Activities

- **Sensor hunt:** Use a free sensor app to display readings from the accelerometer and gyroscope in real time. Tilt, shake, and spin the phone.
- **Compass challenge:** Use the compass app to navigate from one room to another using only compass bearings.
- **Mapping sensors:** Draw a phone outline and annotate where each sensor sits and what it does.

Curriculum Links

Science (forces, magnetism, sound, light), Computing (input devices), Mathematics (measurement, direction).

Chapter 8: Apps: The Workers Inside Your Phone

Discussion Questions

1. What is the difference between a free app and a paid app? Are free apps really free?
2. Why is it important to read what permissions an app asks for before installing it?
3. What would a world without apps look like? What would you miss most?

Extension Activities

- **App audit:** List all apps on a device and categorise them by purpose. Which category has the most?
- **Permission check:** Look at the permissions a familiar app requests. Discuss whether each permission makes sense for what the app does.
- **Design your own app:** Sketch a paper prototype for an app that solves a real problem in your life.

Curriculum Links

Computing (software design, algorithms), PSHE (privacy, digital literacy), Citizenship (consumer awareness).

Chapter 9: GPS: Finding Your Way

Discussion Questions

1. GPS works in remote forests and deserts. Why do you think it works even with no phone signal?
2. How did people navigate before GPS? What tools did they use?
3. Can you think of any problems with always knowing where you are (or others knowing where you are)?

Extension Activities

- **Satellite map:** Look up how many GPS satellites are currently in orbit. Draw their orbits around a globe.
- **Triangulation activity:** Use three circles drawn on a map (each representing the range from a different point) to find where they overlap — that is triangulation.
- **Paper map navigation:** Navigate a short local route using only a printed paper map — no phone allowed.

Curriculum Links

Geography (navigation, mapping, coordinates), Science (satellites, orbits), Mathematics (geometry, measurement).

Chapter 10: Stay Safe and Smart

Discussion Questions

1. Why is a fingerprint or face unlock more secure than a simple number PIN?
2. What personal information should you *never* share in an app or online?
3. What does “being kind online” mean to you? Is it different from being kind in person?

Extension Activities

- **Password strength test:** Use an offline password checker to rate the strength of sample passwords (not real ones).
- **Permission roleplay:** Act out scenarios where an app asks for a suspicious permission — children decide whether to allow or deny.
- **Digital kindness pledge:** Create a class or family digital kindness charter and display it near your devices.

Curriculum Links

PSHE (online safety, digital wellbeing), Computing (cybersecurity basics), Citizenship (rights and responsibilities online).

Connections to Book 1

- **Chapter 1** (What Is a Smartphone?) links to **Book 1, Chapter 1** (What Is the Internet?) — both are entry points into digital technology.
- **Chapter 8** (Apps) links to **Book 1, Chapter 2** (Websites and Browsers) — both explain how software is the layer users interact with.
- **Chapter 9** (GPS) extends **Book 1, Chapter 6** (How Messages and Videos Travel) — data packets vs. satellite signals.
- **Chapter 10** (Stay Safe and Smart) directly continues **Book 1, Chapter 9** (Stay Safe and Curious) with device-specific safety.
- **Consider adding:** A connectivity chapter (Wi-Fi, Bluetooth, 4G/5G) to close the loop with Book 1's Wi-Fi and routers content.

Series: The Secret Life of Technology

Book 1: The Secret Life of the Internet ✓

Book 2: Inside a Smartphone: The Tiny World in Your Hand ✓

Index

- Accelerometer, 27
- accelerometer, 25
- App, 30
- app, 29
- App store, 30
- app store, 30
- Binary code, 7
- binary code, 6
- Biometrics, 37
- calculator, 2
- calendar, 2
- camera, 2
- Camera sensor, 18
- camera sensor, 17
- Capacitive touchscreen, 14
- capacitive touchscreen, 13
- Charging, 22
- Chemical energy, 22
- chemical energy, 21
- Compass, 27
- compass, 27
- computer, 2
- Digital zoom, 18
- Electric current, 22
- electric current, 21
- front camera, 17
- Gesture, 14
- gestures, 13
- Gigabyte (GB), 11
- gigabytes, 10
- Gigahertz, 7
- gigahertz, 7
- GPS, 32, 33
- Gyroscope, 27
- gyroscope, 25
- Input, 2, 4
- Instruction, 7
- instructions, 6
- Internal storage, 10
- Lens, 18
- lens, 16
- Light sensor, 27
- light sensor, 25
- Lithium-ion battery, 22
- lithium-ion battery, 21
- Memory, 2, 4
- Microphone, 27
- microphone, 26
- music player, 2
- navigation device, 2
- Night mode, 17, 18
- Operating system, 11, 30
- operating system, 29
- Optical zoom, 18
- Output, 2, 4
- password, 37
- permissions, 38
- Pixel, 14, 18
- pixels, 14, 17
- Processor, 4, 7
- processor, 2, 6
- proximity sensor, 25
- RAM, 9, 11
- rear camera, 17
- Receiver, 33
- receiver, 33
- Satellite, 33
- Screen resolution, 14
- screen resolution, 14
- Selfie, 18
- Sensor, 27
- Smartphone, 4

Index

Storage, [2](#), [4](#), [11](#)

telephone, [2](#)

torch, [2](#)

Transistor, [7](#)

transistors, [6](#)

Triangulation, [33](#)

triangulation, [33](#)

Update, [30](#)

update, [30](#)

web browser, [2](#)